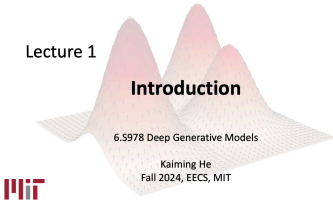
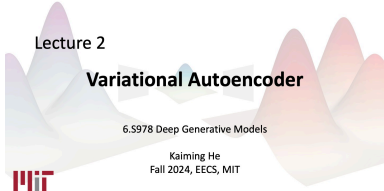
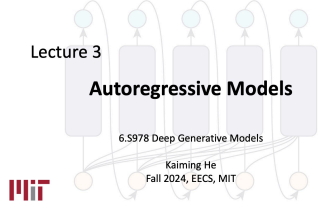
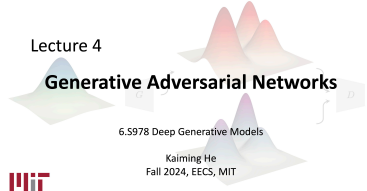


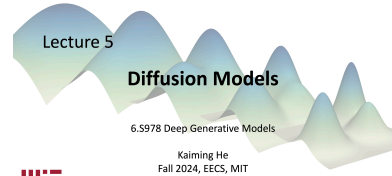
6.S978 Deep Generative Models

MIT EECS, Fall 2024

This schedule is preliminary and subject to change as the term evolves.

Date	Topics	Course Materials
Week 1		
Thurs 09/05/2024	Lecture: Introduction	Lecture 1  slides (assets/pdfs/lec1_intro.pdf)
Week 2		
Tues 09/10/2024	Reading: Modeling Image Prior	Reading List: 1. D. Zoran and Y. Weiss, From Learning Models of Natural Image Patches to Whole Image F 2011 [PDF] (https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=6126278) 2. D. Zoran and Y. Weiss, Natural Images, Gaussian Mixtures and Dead Leaves, NeurIPS 20 (https://proceedings.neurips.cc/paper_files/paper/2012/file/e97ee2054defb209c35fe4dc94! 3. D. Ulyanov et al., Deep Image Prior, CVPR 2018 [PDF] (https://openaccess.thecvf.com/content_cvpr_2018/papers/Ulyanov_Deep_Image_Prior_C (Optional) Recommended readings: 1. A. A. Efros and T. K. Leung, Texture Synthesis by Non-parametric Sampling, 1999 [PDF] (https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=790383) 2. C. Barnes et al., PatchMatch: A Randomized Correspondence Algorithm for Structural Ima (https://3dvar.com/Barnes2009PatchMatch.pdf) 3. W. T. Freeman et al., Example-Based Super-Resolution, 2002 [PDF] (https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=988747)
Thurs 09/12/2024	Lecture: Variational Autoencoder (VAE)	Lecture 2  slides (assets/pdfs/lec2_vae.pdf)
Week 3		
Tues 09/17/2024	Reading: Normalizing Flows	Reading List: 1. D. Rezende and S. Mohamed, Variational Inference with Normalizing Flows, ICML 2015 [P (https://proceedings.mlr.press/v37/rezende15) 2. D. P. Kingma and P. Dhariwal, Natural Images, Glow: Generative Flow with Invertible 1x1 C 2018 [PDF] (https://proceedings.neurips.cc/paper_files/paper/2018/hash/d139db6a236200b21cc7f752 Abstract.html) 3. J. Behrmann et al., Invertible residual networks, ICML 2019 [PDF] (https://proceedings.mlr.press/v97/behrmann19a.html?ref=https://githubhelp.com) (Optional) Recommended readings: 1. Lilian Weng's post on "Flow-based Deep Generative Models" [Link] (https://lilianweng.github.io/13-flow-models/)

Thurs 09/19/2024	Lecture: Autoregressive (AR) Models	 Lecture 3 Autoregressive Models 6.S978 Deep Generative Models Kaiping He Fall 2024, EECS, MIT slides (assets/pdfs/lec3_ar.pdf)
Week 4		
Tues 09/24/2024	Reading: Autoregressive (AR) Models	Reading List: 1. Y. Bengio and S. Bengio, Modeling High-Dimensional Discrete Data with Multi-Layer Neural Networks, 1999 [PDF] (https://proceedings.neurips.cc/paper_files/paper/1999/file/e6384711491713d2) 2. A. Van Den Oord et al., Pixel Recurrent Neural Networks, ICML 2016 [PDF] (https://proceedings.mlr.press/v48/oord16.pdf) 3. D. P. Kingma et al., Improved Variational Inference with Inverse Autoregressive Flow, NIPS 2016 [PDF] (https://proceedings.neurips.cc/paper_files/paper/2016/file/ddeebdeefdb7e7e7a697e1c3e3)
Thurs 09/26/2024	Reading: AR and tokenizers	Reading List: 1. L. Yu et al., Language Model Beats Diffusion -- Tokenizer is Key to Visual Generation, ICLR 2024 [PDF] (https://arxiv.org/abs/2310.05737) 2. K. Tian et al., Visual Autoregressive Modeling: Scalable Image Generation via Next-Scale Image Generation, ICLR 2024 [PDF] (https://arxiv.org/abs/2404.02905) 3. Q. Yu et al., An Image is Worth 32 Tokens for Reconstruction and Generation, arXiv [PDF] (https://arxiv.org/abs/2406.07550) (Optional) Recommended readings: 1. M. Chen et al., Generative Pretraining from Pixels, ICML 2020 [PDF] (https://proceedings.mlr.press/v119/chen20s/chen20s.pdf) 2. A. Ramesh et al., Zero-Shot Text-to-Image Generation (DALL-E1), ICML 2021 [PDF] (https://arxiv.org/pdf/2102.12092) 3. F. Mentzer et al., Finite Scalar Quantization: VQ-VAE Made Simple, arXiv [PDF] (https://arxiv.org/abs/2309.15505)
Week 5		
Tues 10/01/2024	Reading: AR and Diffusion	Reading List: 1. T. Li et al., Autoregressive Image Generation without Vector Quantization, arXiv [PDF] (https://arxiv.org/abs/2406.11838) 2. C. Zhou et al., Transfusion: Predict the Next Token and Diffuse Images with One Multi-Modal Transformer, arXiv [PDF] (https://www.arxiv.org/abs/2408.11039) 3. J. Xie et al., Show-o: One Single Transformer to Unify Multimodal Understanding and Generation, arXiv [PDF] (https://arxiv.org/abs/2408.12528) (Optional) Recommended readings: 1. E. Hoogeboom et al., Autoregressive Diffusion Models, ICLR 2022 [PDF] (https://arxiv.org/abs/2110.08877) 2. B. Chen et al., Diffusion Forcing: Next-token Prediction Meets Full-Sequence Diffusion, arXiv [PDF] (https://arxiv.org/abs/2407.01392)
Thurs 10/03/2024	Lecture: Generative Adversarial Network (GAN)	 Lecture 4 Generative Adversarial Networks 6.S978 Deep Generative Models Kaiping He Fall 2024, EECS, MIT slides (assets/pdfs/lec4_gan.pdf)
Week 6		

Tues 10/08/2024	Reading: GAN 1	Reading List: 1. A. Sauer et al., StyleGAN-T- Unlocking the Power of GANs for Fast Large-Scale Text-to-Img 2023 [PDF] (https://arxiv.org/abs/2301.09515) 2. M. Kang et al., Scaling up GANs for Text-to-Image Synthesis, CVPR 2023 [PDF] (https://arxiv.org/abs/2303.05511) 3. M. Kang et al., Distilling Diffusion Models into Conditional GANs, ECCV 2024 [PDF] (https://arxiv.org/abs/2405.05967) (Optional) Recommended readings: 1. Z. Tu. Learning Generative Models via Discriminative Approaches, CVPR 2007 [PDF] (https://ieeexplore.ieee.org/abstract/document/4270060)
Thurs 10/10/2024	Reading: GAN 2	Reading List: 1. N. Huang et al., The GAN is dead; long live the GAN! A Modern GAN Baseline, ICML 2024 (https://openreview.net/pdf/9197a4074900f763675c3c896d7c4ca3402a3c55.pdf) 2. Z. Wang et al., Diffusion-GAN: Training GANs with Diffusion, ICLR 2023 [PDF] (https://arxiv.org/abs/2306.01654) 3. S. Asokan et al., GANs Settle Scores!, arXiv [PDF] (https://arxiv.org/abs/2306.01654)
Week 7		
Tues 10/15/2024	No class (student holiday)	
Thurs 10/17/2024	Lecture: Energy-based Models, Score matching, Diffusion Models	 slides (assets/pdfs/lec5_diffusion.pdf)
Week 8		
Tues 10/22/2024	Reading: Diffusion Models	Reading List: 1. J. Ho and T. Salimans, Classifier-Free Diffusion Guidance, NeurIPS 2021 [PDF] (https://arxiv.org/abs/2207.12598) 2. T. Salimans and J. Ho, Progressive Distillation for Fast Sampling of Diffusion Models, ICLR (https://arxiv.org/abs/2202.00512) 3. E. Hoogeboom et al., Simple diffusion: End-to-end diffusion for high resolution images, ICM (https://arxiv.org/abs/2301.11093) (Optional) Recommended readings: 1. R. Rombach et al., High-Resolution Image Synthesis with Latent Diffusion Models, CVPR : (https://arxiv.org/abs/2112.10752) 2. T. Karras et al., Elucidating the Design Space of Diffusion-Based Generative Models, Neur (https://arxiv.org/abs/2206.00364) 3. A. Ramesh et al., Hierarchical Text-Conditional Image Generation with CLIP Latents, arXiv (https://arxiv.org/abs/2204.06125) 4. T. Chen, On the Importance of Noise Scheduling for Diffusion Models, arXiv [PDF] (https://arxiv.org/pdf/2301.10972) 5. Sander Dieleman's post on "Diffusion is spectral autoregression" [Link] (https://sander.ai/2023/10/20/diffusion-is-spectral-autoregression.html)

Thurs 10/24/2024	Reading: Diffusion beyond Denoising	Reading List: 1. A. Bansal et al., Cold Diffusion: Inverting Arbitrary Image Transforms Without Noise, NeurIPS (https://arxiv.org/abs/2208.09392) 2. S. Rissanen et al., Generative Modelling With Inverse Heat Dissipation, ICLR 2023 [PDF] (https://arxiv.org/abs/2206.13397) 3. M. Delbracio et al., Inversion by Direct Iteration- An Alternative to Denoising Diffusion for IR TMLR 2023 [PDF] (https://arxiv.org/abs/2303.11435) (Optional) Recommended readings: 1. G. Daras et al., Soft Diffusion: Score Matching for General Corruptions, TMLR 2023 [PDF] (https://arxiv.org/abs/2209.05442)
Week 9		
Tues 10/29/2024	Reading: Discrete Diffusion	Reading List: 1. J. Austin et al., Structured Denoising Diffusion Models in Discrete State-Spaces, NeurIPS (https://arxiv.org/abs/2107.03006) 2. S. Gong et al., DiffuSeq: Sequence to Sequence Text Generation with Diffusion Models, ICML (https://arxiv.org/abs/2210.08933) 3. A. Lou et al., Discrete Diffusion Modeling by Estimating the Ratios of the Data Distribution, ICML (https://arxiv.org/abs/2310.16834)
Thurs 10/31/2024	Reading: Flow Matching 1	Reading List: 1. Y. Lipman et al., Flow Matching for Generative Modeling, ICLR 2023 [PDF] (https://arxiv.org/abs/2210.09850) 2. M. S. Albergo et al., Building Normalizing Flows with Stochastic Interpolants, ICLR 2023 [PDF] (https://arxiv.org/abs/2209.15571) 3. X. Liu et al., Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow, ICLR 2023 [PDF] (https://arxiv.org/abs/2209.03003) (Optional) Recommended readings: 1. T. Fjelde et al., Post on "An Introduction to Flow Matching" [Link] (https://mlg.eng.cam.ac.uk/blog/2024/01/20/flow-matching.html#figure-forward_samples-or)
Week 10		
Tues 11/05/2024	Reading: Flow Matching 2	Reading List: 1. P. Esser et al., Scaling Rectified Flow Transformers for High-Resolution Image Synthesis, ICLR 2024 [PDF] (https://arxiv.org/abs/2403.03206) 2. N. Ma et al., SIT: Exploring Flow and Diffusion-based Generative Models with Scalable Interpolants, ECCV 2024 [PDF] (https://arxiv.org/abs/2401.08740) 3. I. Gat et al., Discrete Flow Matching, arXiv [PDF] (https://arxiv.org/abs/2407.15595)
Thurs 11/07/2024	Guest Lecture: Jun-Yan Zhu https://www.cs.cmu.edu/~junyanz/ - Ensuring Data Ownership in Generative Models	 Ensuring Data Ownership in Generative Visual Models Jun-Yan Zhu Generative Intelligence Lab Carnegie Mellon University slides (assets/pdfs/data_ownership_MIT_guest_lecture.pdf)
Week 11		
Tues 11/12/2024	Reading: Application - Videos	Reading List: 1. O. Bar-Tal et al., Lumiere: A Space-Time Diffusion Model for Video Generation, arXiv [PDF] (https://arxiv.org/abs/2401.12945) 2. J. Bruce et al., Genie: Generative Interactive Environments, arXiv [PDF] (https://arxiv.org/abs/2401.12945) 3. The Movie Gen team @ Meta, Movie Gen: A Cast of Media Foundation Models, arXiv [PDF] (https://ai.meta.com/static-resource/movie-gen-research-paper)

Thurs 11/14/2024	Reading: Application - 3D and Geometry	Reading List: 1. B. Poole et al., DreamFusion: Text-to-3D using 2D Diffusion, ICLR 2023 [PDF] (https://arxiv.org/abs/2311.04400) 2. Y. Hong et al., LRM: Large Reconstruction Model for Single Image to 3D, ICLR 2024 [PDF] (https://arxiv.org/abs/2311.04400) 3. Y. Siddiqui et al., MeshGPT: Generating Triangle Meshes with Decoder-Only Transformers (https://nihalsid.github.io/mesh-gpt/) (Optional) Recommended readings: 1. L. Zhang et al., CLAY: A Controllable Large-scale Generative Model for Creating High-quality SIGGRAPH 2024 [PDF] (https://arxiv.org/abs/2406.13897) 2. X. Wei et al., MeshLRM: Large Reconstruction Model for High-Quality Meshes, arXiv [PDF] (https://sarahweiii.github.io/meshlrn/) 3. T. Shen et al., Flexible Isosurface Extraction for Gradient-Based Mesh Optimization, SIGGRAPH 2024 [PDF] (https://research.nvidia.com/labs/toronto-ai/flexicubes/)
Week 12		
Tues 11/19/2024	Reading: Application - Robotics	Reading List: 1. M. Janner et al., Planning with Diffusion for Flexible Behavior Synthesis, ICML 2022 [PDF] (https://arxiv.org/abs/2205.09991) 2. C. Chi et al., Diffusion Policy: Visuomotor Policy Learning via Action Diffusion, RSS 2023 [PDF] (https://arxiv.org/abs/2303.04137) 3. S. Yang et al., UniSim: Learning Interactive Real-World Simulators, ICLR 2024 [PDF] (https://arxiv.org/abs/2310.06114) (Optional) Recommended readings: 1. D. Driess et al., PaLM-E: An Embodied Multimodal Language Model, ICML 2023 [PDF] (https://arxiv.org/abs/2303.03378)
Thurs 11/21/2024	Guest Lecture: Yang Song (https://yang-song.net/) - Consistency Models	Consistency Models Yang Song  Prafulla Dhariwal Mark Chen Ilya Sutskever Cheng Lu slides (assets/pdfs/CM_lecture.pdf)
Week 13		
Tues 11/26/2024	Reading: Application - Material Science	Reading List: 1. W. Jin et al., Junction Tree Variational Autoencoder for Molecular Graph Generation, ICML 2022 [PDF] (https://arxiv.org/pdf/1802.04364) 2. E. Hoogeboom et al., Equivariant Diffusion for Molecule Generation in 3D, ICML 2022 [PDF] (https://proceedings.mlr.press/v162/hoogeboom22a) 3. G. Zhou et al., Uni-Mol: A Universal 3D Molecular Representation Learning Framework, ICML 2022 [PDF] (https://openreview.net/forum?id=6K2RM6wVqKu) (Optional) Recommended readings: 1. M. Xu et al., Geometric Latent Diffusion Models for 3D Molecule Generation, ICML 2023 [PDF] (https://arxiv.org/abs/2305.01140) 2. M. Arts et al., Two for One: Diffusion Models and Force Fields for Coarse-Grained Molecules, ICML 2023 [PDF] (https://arxiv.org/abs/2302.00600)
Thurs 11/28/2024	No class (Thanksgiving)	
Week 14		

Tues 12/03/2024	Reading: Application - Protein and Biology	Reading List: 1. J. L. Watson et al., De novo design of protein structure and function with RFdiffusion, Nature (https://www.nature.com/articles/s41586-023-06415-8) 2. J. Abramson et al., Accurate structure prediction of biomolecular interactions with AlphaFold (https://www.nature.com/articles/s41586-024-07487-w) 3. J. B. Ingraham et al., Illuminating protein space with a programmable generative model, Nature (https://www.nature.com/articles/s41586-023-06728-8)
Thurs 12/05/2024	Final Presentation 1	
Week 15		
Tues 12/10/2024	Final Presentation 2	